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Government political structure and violent death rates: A longitudinal analysis of forty-three countries, 1960–2008

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Abstract

Objectives: Currently, little is known regarding the effect of regime type on mortality on a global level. The purpose of this study is to examine the effect of regime type on the rates of violent deaths (homicide, suicide, and combined rates).

Methods: Three measures of democracy were used to quantify regime type, the independent variable. Homicide and suicide rates were obtained from the World Health Organization. Multivariate conditional fixed-effects models were run to examine associations between regime characteristics and logged rates of homicide, suicide, and violent deaths. Models were adjusted for unemployment and economic inequality.

Results: Nations that scored higher on democracy indices, especially emerging democracies, experienced increased mortality due to violence. Homicide and suicide were divergent, showing a different time course and decreasing statistical power as a combined variable. Unemployment and inequality were associated with higher violence-related mortality.

Conclusions: Homicide and suicide appear to be more prevalent in democracies. Future analyses should examine which aspects of democracies lead to higher rates of violent death and should seek to use independently collected mortality data.

Keywords

Homicide; Suicide; Violence; Democracy; Regime

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Research involving human participants and/or animals
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N/A.

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None.

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1. Introduction

Since the 49th World Health Assembly (WHA) declared violence a public health priority (WHA, 1996), violence has received widespread attention as a preventable human problem. This means that vast reductions in many illnesses through public health campaigns are also achievable with various forms of violence (Krug et al., 2002). Indeed, evidence points to the highly influential nature of social, economic, and even political contributors to violence (Brownstein, 2000; Hibbs, 1973; Lee, Lee, & Marotta, 2018; Miguel, Satyanath, & Sergenti, 2004). While biological, psychological, and other immediate factors determine the possibility of violence in a given individual, sociocultural, economic, and political factors have the power to elicit or eliminate epidemics of violence (Gilligan, Lee, Garg, Blay-Tofey, & Luo, 2016).

As a consequence, violence rates fluctuate over time and, during times of great changes, fluctuate greatly. After a period of worldwide upheaval, violent death rates decreased slightly between, 1990 and 2010, with age-standardized rate reductions by 1% for homicide and by 9.6% for suicide (Lozano et al., 2012). There are many advantages to considering overall violent death rates for assessment at a larger scale; for instance, while homicides and suicides often rise or fall together, nations with similar violent death rates may show lower homicide rates but higher suicide rates for some, and higher homicide rates but lower suicide rates for others (Lee, Marotta, et al., 2014). Additionally, considering one type alone can give false assurance that violence rates are falling, when they may simply be transitioning from the battlefield into the home, or from being directed against others toward against the self, as cultures shift and country conditions change (Krug et al., 2002). It is also important to note that, although homicides have decreased since their peak twenty years ago, both homicides and suicides now account for a greater proportion of the global burden of disease (WHO, 2017a), indicating that progress has been slower than in other illnesses. In 2012, suicide accounted for 1.4% of deaths worldwide (50% of all violent deaths in men and 71% in women), causing the WHO to declare suicide-related mortality reduction as a global imperative (Lozano et al., 2012; Turecki and Brent, 2016; WHO, 2014).

Violent death rates can be seen as a barometer for the health of a society (Lee, Wexler, & Gilligan, 2014; Wilkinson & Pickett, 2010). There is growing evidence of the direct role that social policies and political structures play in mortality, especially in violent deaths (Lee, 2019; Lozano et al., 2012). Violent deaths have been associated with greater unemployment and higher Gini coefficients (Andres, 2005; Jacobs & Richardson, 2008; McCall, Parker, & MacDonald, 2008). The ecological model of violence provides an understanding of the risk factors from an individual to a societal viewpoint (Gilligan, 2000; Adshead et al., 2013). Existing evidence shows that level of governance, or strength of rule of law, is negatively correlated with homicide rates (Cole and Gramajo, 2009). However, governance and level of democratization are conceptually distinct, and currently there is little published on the effect of regime type on mortality on a global level (Gilligan et al., 2016). We know that transitional democracies have the highest violent crime rates of any regime type (Altman & Pérez-Liñán, 2002; LaFree and Tseloni, 2006; Pérez-Liñán and Mainwaring, 2013). This study examined the association between democratization and rates of suicide, homicide, and violent deaths from 1960 to 2008 in 43 nations.

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2. Methods

2.1. Sample

A sample of countries was selected from 75 member nations included in the WHO dataset of homicide and suicide mortalities. Countries providing fewer than five years of data were excluded, leaving a final dataset of 43 countries: Argentina, Australia, Austria, Belarus, Belgium, Brazil, Bulgaria, Canada, Chile, Colombia, Costa Rica, Denmark, Estonia, Finland, France, Georgia, Greece, Guatemala, Hungary, Ireland, Israel, Italy, Japan, Kazakhstan, Kyrgyzstan, Mexico, Netherlands, New Zealand, Norway, Panama, Poland, Portugal, Panama, Republic of Korea, Singapore, Spain, Sweden, Switzerland, Ukraine, United Kingdom, United States of America, Uruguay, and Venezuela.

2.2. Data

Data for this study come from several open-access global repositories of information.

Dependent variables: Suicide and homicide mortality rates.—The WHO compiles mortality data reported annually through the civil-registration systems of Member States (WHO, 2017a). Historical-series data consisting of age-standardized death rates by age group and sex were obtained through the WHO Violence and Injury Prevention Division repository of data (WHO, 2017b). The WHO calculated age-standardized rates of mortalities from suicide and homicide of 75 nations from 1950 to 2008 using ICD-7, 8, 9 and 10 codes with World Standard Population Adjustment (Ahmad et al., 2001) for nations with data completeness > 75% and populations > 500,000 (WHO, 2017b).

Primary independent variable: Democracy.—The study extracted information regarding multiple dimensions of democratization from three sources: the Polity IV Project, Freedom House, and Polyarchy datasets.

The *Polity Project IV: Political Regime Characteristics and Transitions*, *1800–2015* dataset examines the extent to which states are autocratic, for cross-national political inquiry and quantitative analysis (Center for Systemic Peace, 2017). For this analysis, the years 1960–2008 were studied. Polity conceptualizes democracy along three dimensions: 1) policies and institutions that allow citizens to voice alternative perspectives and support competing leadership, or “Openness and competitiveness of executive recruitment”; 2) procedures and policies that temper the executive branch’s power, or “Constraints on the chief executive”; and 3) protections for political participation through a system of civil liberties, or “Political participation” (Marshall & Jaggers, 2002; Marshall, Jaggers, & Gurr, 2016). These dimensions comprise the Polity scale, a 21-point indicator of regime authoritativeness that ranges from −10 (hereditary monarchy) to +10 (consolidated democracy). Based on the Polity score, three broad variables were created to indicate scores of −10 to −6 as autocratic, −5 to 5 as anocratic and 6 to 10 as democratic regimes (Marshall, Jaggers, & Gurr, 2016).

Freedom House’s *Annual Comparative Survey of Freedom from 1972 to 2012* operationalizes democracy through a focus on political and civil rights (Freedom House, 2016a, b). For this study the years 1972–2008 were analyzed. Countries are assigned ratings, 1 through 7, on civil liberties and on political rights, with 7 indicating the least freedom and

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1 indicating the greatest freedom, based on their scores on ten political-rights and fifteen civil-liberties questions (Freedom House, 2014). The mean of the civil-liberties and political-rights subscores (1–7) provides an overall rating of freedom (Freedom Rating), which forms the basis of the classification of countries as Free (1.0–2.5), Partly Free (3.0–5.0), or Not Free (5.5–7.0). Three dichotomous dummy variables were created using political-rights and civil-liberties scores to indicate whether countries were Free, Partly Free, or Not Free.

The Polyarchy dataset uses electoral data to operationalize democracy on the basis of two core theoretical constructs: (1) competition within political institutions; and (2) participation in the process of governance (Dahl, 1973; Vanhanen, 1990, 1997, 2000a). This study analysis focused on 1960–2000. Competition uses the share of votes cast for non-governing parties during elections, to measure opportunities for the public to gain control over political institutions (Vanhanen, 2003, 2004). Participation reflects the degree to which people engage in politics by measuring the percentage of the population that voted in elections (Vanhanen, 2003, 2004). The Index of Democracy is the product of Competition and Participation divided by 100, with higher scores indicating greater democratization (Altman & Pérez-Liñán, 2002; Vanhanen, 1990, 1997, 2000c, 2003, 2004).

Independent variable – covariates: Economic inequality and unemployment.— Gini coefficients were obtained from the Gini Database (Milanovic and van der Weide, 2014; Milanovic, 2016) distributed by the World Bank and the UN University Database of Gini Coefficients (UNU-WIDER, 2017). The Gini database consists of national estimates using household surveys compiled from seven databases of economic data. The WIDER World Income Inequality Database, Version 2.0c, May 2008, retrieved Gini coefficients for inclusion in the analysis. Unemployment data included annual World Bank estimates of percentage unemployed as a proportion of the national population.

2.3. Statistical analyses

The current study uses a panel dataset of variables compiled from several sources to elucidate relationships between democratization and rates of homicide, suicide, and overall violent deaths in a sample of 43 nations from the period 1960–2008. Raw data files were cleaned, imported, and merged into a single file for analysis (StataCorp, 2015). All data-merging and analysis used Stata v. 14 xtreg commands with a fixed-effects estimator and year dummy variables (Cameron & Trivedi, 2010). Multilevel cross-national data in which observations are clustered within higher-order countries demand statistical models that correctly handle the hierarchical structure of the data (Goldstein, 2005; Rabe-Hesketh & Skrondal, 2008). Fixed-effects analyses reduce omitted-variable bias by controlling for time-invariant characteristics (Rabe-Hesketh & Skrondal, 2008). Rates of homicide, suicide, and violent deaths were transformed to the logarithm scale before fixed-effects Ordinary Least Squares regression analyses were performed. Logarithmic transformation linearizes highly skewed data to more closely resemble a normal distribution (Benoit, 2011). In this study, a log-level regression specification was performed in which natural log-transformed values of mortality rates were regressed on raw values of democratization. Log-transformed coefficients can be interpreted thus: a change of 1% in the outcome variable on the log scale

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corresponds to a one-unit change (+/−) in the independent variable. The log-log specification implies interpretation as change of 1% in logged unemployment rates on percent change of logged rates of mortality. Robust Huber-White standard errors using asymptotic maximum like lihood estimation provided a diagnostic tool to indicate model stability (Maas & Hox, 2004) and to correct the variance for the presence of potential serial autocorrelation and heteroscedasticity (Cameron & Miller, 2015; Hoechle, 2007; Stock & Watson, 2008).

Fixed-effects statistical models were run to examine associations between democratization variables and logged rates of homicide, suicide, and violent deaths, overall and stratified by men and women. Continuous democratization variables included: 1) Polity IV score; 2) Freedom House democracy score; and 3) the Index of Democracy from the Polyarchy dataset. Data are presented for base models containing only democratization variables and models adjusting for Gini coefficients and unemployment rates (log). Significance tests are presented for both unstandardized and cluster-adjusted Huber-White standard errors at the $p < 0.05$ and $p < 0.1$ thresholds.

3. Results

3.1. Descriptive results

Table 1 provides descriptive statistics for dependent variables (homicide, suicide, and overall death rates) and independent variables (democratization and inequality variables), overall and for each country in the sample. Violent death rates were 18.33 per 100,000 overall, with a mean age-standardized suicide rate of 12.11 per 100,000 (SD = 7.15) and a mean age-standardized homicide rate of 6.22 per 100,000 (SD = 10.74).

3.1.1. Polity IV project—Through the interval of 1960–2008, the mean of the Polity scores was 7.07 (SD = 5.40), with autocracy and democracy scores of 0.89 (2.20) and 7.99 (3.30) respectively. Nearly 85% of the observations were in democratic regimes (83.53%, n = 710) with the remaining almost evenly split between autocratic (6.82%, n = 58) and anocratic (6.82%, n = 58).

3.1.2. Freedom house—Throughout the study period of 1972–2008, Free states accounted for the greatest proportion of observations with 81.49% (n = 546), followed by 13.73% (n = 92) Partly Free and 4.78% (n = 32) Not Free. The mean freedom house score was 2.16 (SD = 1.62).

3.1.3. Polyarchy—Over 1960–2000, the mean Index of Democracy was 24.26 (SD = 12.84).

3.2. Multivariate conditional year and country fixed-effects analyses

3.2.1. Polity IV project—Table 2 presents base and adjusted models of fixed-effects regressions of associations between polity score and average logged rates of homicide, suicide, and violent deaths from a period of 1960–2008. In the adjusted models, greater democracy measured by the polity scale was significantly associated with greater rates of suicide death rates ($\beta = 0.01$, $SE_{HW} = 0.005$, $p < 0.01$). Unemployment rates (logged) were

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associated with greater rates of homicide ($\beta = 0.09$, $SE_{HW} = 0.03$, $p < 0.05$), suicide ($\beta = 0.06$, $SE_{HW} = 0.03$, $p < 0.05$), and violent ($\beta = 0.11$, $SE_{HW} = 0.03$, $p < 0.05$) death rates.

3.2.2. Freedom house—Fixed-effects regressions of associations between democracy and rates of homicide are presented in Table 3. Freedom house score measuring greater autocracy was associated with lower rates of suicide ($\beta = -0.05$, $SE_{HW} = 0.02$, $p < 0.05$). Unemployment was significantly associated with increased logged homicide ($\beta = 0.14$, SE = 0.05, $p < 0.05$) suicide ($\beta = 0.06$, $SE_{HW} = 0.06$, $p < 0.05$) and violent death ($\beta = 0.09$, $SE_{HW} = 0.04$, $p < 0.05$) rates.

3.2.3. Polyarchy—Table 4 presents fixed-effects regression analyses of associations between the Index of Democracy, and rates of homicide and suicide from 1968 to 2000. Higher scores on the Index of Democracy was associated with greater logged rates of violent deaths ($\beta = 0.007$, $SE_{HW} = 0.003$, $p < 0.05$, $SE_{HW} = 0.001$, $p < 0.05$). Scores on the Index of Democracy were associated with greater rates of suicide ($\beta = 0.007$, $SE_{HW} = 0.002$, p <6 0.05). A similar trend occurred in which higher scores on the Index of Democracy scale were associated with lower rates of homicide ($\beta = 0.007$, $SE_{HW} = 0.002$, $p < 0.05$). Greater rates of unemployment was associated with increased rates of violent deaths ($\beta = 0.06$, $SE_{HW} = 0.02$, $p < 0.05$).

4. Discussion

In 1974, thirty-five nations were classified as democracies; forty years later, by 2014, the number nearly tripled, to ninety-five. One might have predicted, with the reduction of state oppression and the increase of political equality and economic opportunity, that health levels of societies would increase and violent deaths decrease. While we know that new democracies see spikes in violence levels, regime transitions are fundamentally disruptive, and our expectation is that such nations would stabilize over time. However, human violence is complex, and our study again reveals that answers are not simple. Our results raise important questions about why democracy correlates with greater violence, through its creation and even after the formation of a stable state.

The spike in interpersonal violence as authoritarian repression relaxes during early periods of democratic transition is well-documented in the literature. Neumayer (2003) suggested that democratization and homicide have a nonlinear relationship, in which countries transitioning from autocracies experience a surge in violence. LaFree also illustrated that democracy and crime possess an inverted-U relationship, with a spike in crime during the transition period between autocracy and democracy (LaFree and Tseloni, 2006). By using multiple indices of democracy, we created a comprehensive analysis of levels of democracy (Freedom House, 2016a, b; Marshall & Jaggers, 2002; Vanhanen, 2003).

Freedom for political, social, and cultural groups to organize, to engage in legal processes that oppose the government, and to become parts of political institutions is a defining feature of democratization. These processes characterize healthy democracies and are expected eventually to lessen violence. Hypotheses that the allowance of antagonistic discourse

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encourages intergroup aggression and that the decrease in state authority to police disruptive social currents leads to weaker deterrence for crime apply to some degree.

However, in interpreting the exacerbating effect of democratization on violent death rates, it is important to note that, by virtue of available data, our analysis is skewed toward more developed and more democratic countries. Therefore, the results are likely not generalizable beyond middle and high income countries. Also, authoritarian regimes have an incentive to underreport their violent death rates and it is likely state-sponsored violence may not be captured in the rates of violent death. The data was not adjusted for variance in reporting systems across countries and there is certainly possible measurement error secondary to variance in reporting structures and systems. As well, there is currently little that quantify state-sponsored violence. Meanwhile, it is notable that autocratic countries are more likely to go to war, thus being predisposed to war deaths. Additionally, as the study is longitudinal it measures associations between changes in years and therefore precludes causal inference. The overall time frame captured in this data is relatively short and therefore may not have captured changes in rates that were caused by trends beyond the time frame of this data. Finally, level of governance and the strength of rule of law are correlated to homicide rates (Cole and Gramajo, 2009) but this analysis did not adjust for this variable.

Three other considerations are worth mentioning. The first is the soundness of the democracy indices we used, which are still under development (Giannone, 2010; Sanín et al., 2013; Wejnert, 2014). Secondly, the WHO has not yet standardized mortality data beyond 2008, and while we attempted to standardize them ourselves, too many considerations did not allow us to proceed, effectively leaving out a decade's worth of information in a three decade-long movement of global democratization. A final limitation of this study is that it did not examine any markers of social disorganization which other studies have demonstrated relate to higher rates of violent deaths (Andres, 2005; Jacobs & Richardson, 2008; McCall et al., 2008; Gilligan, 2000; Adshead et al., 2013) beyond the political sphere.

A limitation is that, even though this study is longitudinal, it still measures association between changes in years and precludes causal inference. The sampling of the country is disproportionately from middle- and high-income countries, which biases the sample, and generalizability should be confined to just middle- and high-income countries as a result. Also, we looked at a relatively short period of time, which may not capture changes in rates that were caused by trends prior to the time frame of our data. Finally, our study did not adjust for variance in reporting systems across countries; there is a possibility of some measurement error due to variance in the reporting by the countries. Other limits include: (a) the rates of violent deaths do not take into account the deaths due to state-sponsored violence and/or war (the uncertainty of these measurements caused us to exclude them altogether); (b) the level of governance or strength of rule of law is also correlated with homicide rates, but the analysis did not adjust for this variable; and (c) "markers of social disorganisation" in the literature may be better described to include violent crime.

It is not enough for a country to develop into a democracy. A country must also examine its democratic structure and the durability of its systems, in order to lower rates of violent

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death. For example, some systems may be democratic in name but foster class distinctions and inhibit economic mobility in ways that potently stimulate violence. For new democracies, the system may allow greater room for the outward presentation of discontent, competition, and social discord, especially in times of transition and unmet expectations (Acemoglu, Naidu, Restrepo, & Robinson, 2013; Balcázar, 2016; Bonica, McCarty, Poole, & Rosenthal, 2013). Future analyses can assess these finer aspects of democracy for greater precision. Future directions could also include comparisons with more reliable and independent global databases that do not rely on government reports of death rates. Increased freedom allowed by democracy encompasses the ability for people to pursue their own interests, gain a sense of dignity, and self-regulate in order to self-rule. Such a process may take longer for newer democracies and pose challenges for older ones to maintain their integrity. Additional work might focus on what governments can do to prevent violence within their respective structures as political systems continue their evolution.

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Table 1

Descriptive statistics of observations used in the dataset.

Dependent Variables	**Mean(SD)**	**Range**	**obs = 850**	**1960–2008**	**n = 46**
Suicide	12.11(7.15)	(1.01–38.58)			
Homicide	6.22(10.74)	(.14–90.93)			
Violent deaths	18.33(11.93)	(3.33–94.46)			
Polity IV. Project	**Mean(SD)**	**Range**	**obs = 850**	**1960–2008**	**n = 46**
Polity Score (−10 to +10)	7.07(5.40)	(−9 to +10)			
Autocracy Score (+0 to +10)	.89(2.20)	(+0 to +9)			
Democracy Score (+0 to +10)	7.99(3.30)	(+0 to +10)			
Regime Type	%(n)				
Autocracy	9.41(80)				
Anocracy	6.82(58)				
Democracy	83.53(710)				
Freedom House	**%(n)**		**obs = 726**	**1972–2008**	**n = 43**
Freedom index	2.16(1.62)				
Free	81.49(546)				
Part Free	13.73(92)				
Not Free	4.78(32)				
Polyarchy	**Mean(SD)**	**Range**	**obs = 605**	**1960–2008**	**n = 44**
Index of Democracy (0–50)	24.26(12.84)	(0–47.08)			
Economic inequality	**Mean(SD)**	**Range**	**obs = 722**	**1960–2008**	**n = 46**
Gini Coefficient	37.09(9.29)	(20.20–62.5)			
Unemployment Rate	7.57(4.34)	(.2–24.21)			

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Table 2

Fixed effects regressions of associations between democracy and rates of homicide stratified by age and sex 1950–2008 (obs = 707) n = 43.

	Violent Deaths n = 43		Suicide n = 43		Homicide n = 46	
	Base Model β(SE) [SE_{HW}]	**Adjust. Model (SE)[SE_{HW}]**	**Base Model (SE) [SE_{HW}]**	**Adjust. Model β(SE)[SE_{HW}]**	**Base Model β(SE) [SE_{HW}]**	**Adjust. Model β(SE)[SE_{HW}]**
Polity	.002(.002)[.005]	.01(.003)***[.008]	.01(.002)*[.01]	.01(.003)***[.005]**	.01(.003)*[.01]	.01(.005)*[.01]
Economic covariates						
Gini Coefficient		.003(.002)[.004]		.004(.002)*[.004]		.001(.003)[.001]
UR(Log)		.09(.02)***[.03]*		.06(.03)***[.03]*		.11(.03)***[.05]*

* $p < .05$;

◆ $p < .1$;

Year fixed effects were entered in fixed effects estimation as dummy variables (Not shown in table).

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Table 3

Fixed effects regressions of associations between Freedom House measures of democratization and rates of homicide stratified by age and sex 1972–2008 (obs = 655) n = 43.

	Homicide n = 46		Suicide n = 43		Violent Deaths n = 43	
	Base Model β(SE) [SE_{HW}]	**Adjust.Model β(SE) SE_{HW}]**	**Base Model β(SE) [SE_{HW}]**	**Adjust. Model β(SE) [SE_{HW}]**	**Base Model β(SE) [SE_{HW}]**	**Adjust. Model β(SE)[SE_{HW}]**
Freedom house score	−.02(.01)*[.03]	.02(.02)[.07]	−.02(.008)*[.02]	−.05(.01)***[.02]*	.006(.008)[.02]	.008(.01)[.05]
Economic covariates						
Gini		−.003(.003)[.007]		.005(.002)*[.004]		.003(.002)[.004]
UR(Log)		.14***(.03)[.05]*		.06(.02)***[.06]*		.09(.02)***[.04]*

* $p < .05$;

◆ $p < .1$;

Year fixed effects were entered in fixed effects estimation as dummy variables (Not shown in table).

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Table 4

Fixed effects regressions of association between index of democracy and rates of suicide, homicide and violent deaths (logged) n=44).

	Homicide		Suicide		Violent Death	
	Base Model β(SE) [SE_{HW}]	**Adjust. Model β(SE) [SE_{HW}]**	**Base Model β(SE) [SE_{HW}]**	**Adjust. Model β(SE) [SE_{HW}**	**Base Model β(SE) [SE_{HW}]**	**Adjust. Model β(SE) [SE_{HW}]**
Model 1:						
Index of Democracy	.01(.002)***[.003]	.007(.002)**[.003] *	.005(.001)***[.002] *	.007(.001)***[.002]**	.004(.001)***[.002]	.007(.001)***[.002]**
Economic inequality						
Gini		.004(.003)[.005]		.001(.002)[.003]		.002(.002)[.003]
Unemployment		.09(.03)***[.05]		.05(.01)**[.03]		.06(.01)***[.02] *

* p < .05;

wp < .1;

Models include year and country fixed effects, year fixed effects were entered in fixed effects estimation as dummy variables (Not shown in table).